

Radian Measure and Arc Length



Converting between degrees and radians

1 Convert each angle to radians, in exact form:

a 45°

b 120°

c 270°

d 330°

2 Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6}$

b $\frac{5\pi}{4}$

c $\frac{7\pi}{3}$

d 3.5 rad

Arc length and sector area

3 A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.

4 A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.

5 A wheel of radius 30 cm rolls along the ground without slipping.

a Through what angle, in radians, does the wheel turn when it travels 1.5 m?

b Express this angle to the nearest tenth of a degree.

6 A pendulum of length 80 cm swings through an angle of 25° . Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.